



Institute of Business Administration, Karachi

CSE475 — Compiler Construction — Fall 2026

LEXICAL ANALYZER

Compiler Project Assignment

Released	Sunday, September 13th, 2026	Total Marks	10
Due	Sunday, September 20th, 2026	Submission	LMS
Deliverable	Lexical analyzer + token set	Method	DFA transition table

Assignment

As part of your compiler project, you are required to implement and submit the **Lexical Analyzer (Scanner)** for your proposed programming language. Your scanner must be consistent with the lexical specification of your language and must generate a correct stream of tokens from an input source program.

What to Submit

1. Token Set / Token Specification

- Submit the complete set of tokens supported by your language.
- Clearly identify the **class part** and **value part** where applicable.
- Your token set should include, as relevant: keywords, identifiers, literals, operators, punctuators/separators, comments, and any language-specific tokens.

2. DFA Transition Table

- Submit the transition table used by your lexical analyzer.
- Clearly indicate states, input character classes, transitions, accepting/final states, and error or dead states where applicable.

3. Lexical Analyzer

- Submit a working implementation of the lexical analyzer for your language.
- It must read a source program and generate the corresponding token stream.
- Invalid or unrecognized lexemes must be detected and reported appropriately.

4. Test Input

- Include at least one representative source-code file written in your proposed language.

5. Generated Token Stream

- Include the output produced by your lexical analyzer for the submitted test input.

Implementation Requirement

The lexical analyzer must be implemented using a **DFA transition table**. The transition table should drive the movement from one state to another based on the current input character or character class.

The scanner should not rely primarily on long, hard-coded chains of `if/else` or `switch` statements to recognize individual tokens. Such an implementation will not receive full marks for the transition-table implementation component.

Expected Output Format

For each recognized token, clearly display its **class part** and **value part**. For example:

```
<data_type, float>
<identifier, total>
<assignment_operator, =>
<integer_literal, 25>
```

Submission Checklist

Before uploading to the LMS, confirm that your submission includes:

- Complete token set / token specification.
- DFA transition table.
- Source code for the lexical analyzer.
- Representative test input file(s).
- Generated token stream / output for the submitted test input.
- All files needed to compile and run your scanner.

Important

Your lexical analyzer must reflect the **lexical specification of the language proposed by your group**. The token set, transition table, scanner implementation, and generated output must therefore be mutually consistent.

Submit all required files on the LMS by **Sunday, September 20th, 2026**.

— End of Assignment —